More on PCA

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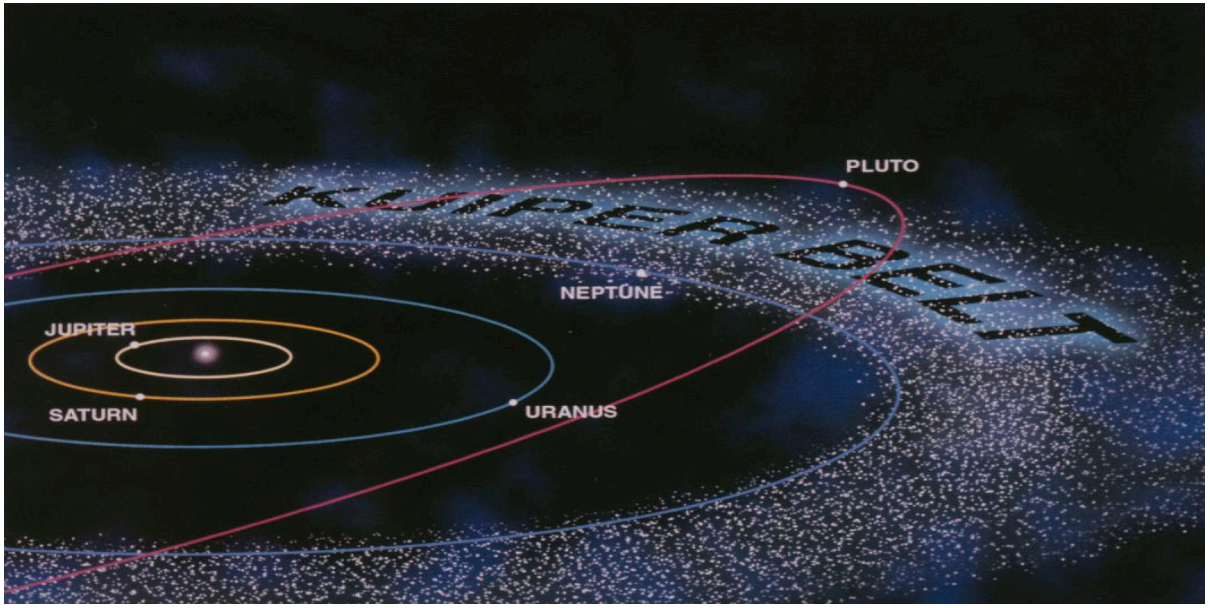
Source/Reading:

- These slides have `R` codes and commentary and in addition the following sections are heavily recommended:

> Chapter 8.1 - 8.4 from Zelterman's book.
>
> Chapter 3.1 - 3.12 from Everitt's book.

Example: Pluto (Zelterman)

- In 2006, Pluto was demoted from the status of a planet and is now considered one of the many objects orbitting the sun at a similar distance.
- KBO: Kuiper belt objects: more than a thousand known objects, and there are estimates of many tens of thousands more yet to be discovered.



Example: Pluto

- Data from the largest KBO's below.
- `Albedo`: light reflected rather than absorbed.
- `Absolute magnitude`: Apparent brightness, corrected for distance in log-scale.
- Is Pluto similar to the other KBOs?

Table 1.3 Several Kuiper Belt objects

Designation		Absolute magnitude	Albedo (%)	Equatorial diameter (km)	Semi-major axis (AU)	Year of discovery
Permanent	Provisional					
Pluto		−1.00	60.0	2320.0	39.4	1930
Makemake	2005 FY9	−0.30	80.0	1500.0	45.7	2005
Haumea	2003 EL61	0.10	84.0	1150.0	43.3	2005
Charon	S/1978 P 1	1.00	40.0	1205.0	39.4	1978
Orcus	2004 DW	2.30	19.7	946.3	39.4	2004
Quaoar	2002 LM60	2.60	19.9	844.0	43.5	2002
Ixion	2001 KX76	3.20	12.0	650.0	39.6	2001
55636	2002 TX300	3.30	10.0	800.0	43.1	2002
55565	2002 AW197	3.30	11.7	734.6	47.4	2002
55637	2002 UX25	3.60	11.5	681.2	42.5	2002
Varuna	2000 WR106	3.70	16.0	500.0	43.0	2000
	2002 MS4	3.80	8.4	726.2	41.8	2002
	2003 AZ84	3.90	12.3	685.8	39.6	2003

Source Wikipedia

Kuiper Belt Object

```
KBO <- read.csv("kuiper_belt_objects.csv")
str(KBO)
```

```
## 'data.frame':    13 obs. of  7 variables:
##  $ Permanent               : chr  "Pluto" "Makemake" "Haumea" "Charon" ...
##  $ Provisional             : chr  "" "2005 FY9" "2003 EL61" "S/1978 P 1" .
##  $ Absolute.magnitude      : num  -1 -0.3 0.1 1 2.3 2.6 3.2 3.3 3.3 3.6 ..
##  $ Albedo....              : num  60 80 84 40 19.7 19.9 12 10 11.7 11.5 ..
##  $ Equatorial.diameter..km.: num  2320 1500 1150 1205 946 ...
##  $ Semi.major.axis..AU.    : num  39.4 45.7 43.3 39.4 39.4 43.5 39.6 43.1
##  $ Year.of.discovery       : int  1930 2005 2005 1978 2004 2002 2001 2002
```

```
Kuiper <- KBO[,-c(1,2)]
colnames(Kuiper) <- c("mag","albedo","diameter","axis","year")
head(Kuiper, n = 3)
```

```
##    mag albedo diameter axis year
## 1 -1.0     60     2320 39.4 1930
## 2 -0.3     80     1500 45.7 2005
## 3  0.1     84     1150 43.3 2005
```

Principal Components

- Perform PCA on the `Kuiper` dataset.

```
KuiPC <- princomp(Kuiper)
summary(KuiPC)
```

```
## Importance of components:
##                             Comp.1       Comp.2       Comp.3       Comp.4
## Standard deviation     467.8572498 20.201186318 7.1607984150 2.139462e+00
## Proportion of Variance   0.9978846  0.001860404 0.0002337634 2.086714e-05
## Cumulative Proportion    0.9978846  0.999745036 0.9999787991 9.999997e-01
##                              Comp.5
## Standard deviation     2.705783e-01
## Proportion of Variance 3.337641e-07
## Cumulative Proportion  1.000000e+00
```

- From this,we see 99.8% of the variability is explained in the first principal component. On the surface, this appears to be a remarkable reduction in the dimensionality of the problem.

Loadings

- The first PC is simply the diameter variable. Why is that?

```
KuiPC$loadings
```

```
##
## Loadings:
##          Comp.1 Comp.2 Comp.3 Comp.4 Comp.5
## mag                                   0.999
## albedo          -0.876 -0.480
## diameter -0.999
## axis                            0.995
## year            -0.478  0.872
##
##                Comp.1 Comp.2 Comp.3 Comp.4 Comp.5
## SS loadings       1.0    1.0    1.0    1.0    1.0
## Proportion Var    0.2    0.2    0.2    0.2    0.2
## Cumulative Var    0.2    0.4    0.6    0.8    1.0
```

Look at the variances

- This happens because the variances (or the standard deviations) of the variables are widely different.
- The `diameter` variable has the largest variability, by far. Hence it dominates the first PC.

```
sapply(Kuiper, sd)
```

```
##        mag     albedo   diameter       axis       year
##   1.720651  27.385416 486.254960   2.595953  20.746331
```

PCA on Correlation matrix

- A better approach, then, is to record all of these values on the same scale. The principal components should be obtained from the correlation matrix in most cases.

- use the `cor=TRUE` option as in this dialog:

```
KuiPCc <- princomp(Kuiper, cor = TRUE)

summary(KuiPCc)
```

```
## Importance of components:
##                           Comp.1    Comp.2     Comp.3      Comp.4      Comp
## Standard deviation     1.786893 1.1359476 0.67202742 0.219707937 0.1294048
## Proportion of Variance 0.638597 0.2580754 0.09032417 0.009654316 0.0033491
## Cumulative Proportion  0.638597 0.8966724 0.98699656 0.996650875 1.0000000
```

- The first two principal components account for almost 90% of the variability.

New Loadings

```
KuiPCc$loadings
```

```
##
## Loadings:
##          Comp.1 Comp.2 Comp.3 Comp.4 Comp.5
## mag       0.538  0.194  0.188  0.161  0.782
## albedo   -0.463 -0.405 -0.446  0.493  0.424
## diameter -0.546         0.216 -0.671  0.451
## axis      0.109 -0.785  0.608
## year      0.432 -0.424 -0.591 -0.529
##
##                Comp.1 Comp.2 Comp.3 Comp.4 Comp.5
## SS loadings       1.0    1.0    1.0    1.0    1.0
## Proportion Var    0.2    0.2    0.2    0.2    0.2
## Cumulative Var    0.2    0.4    0.6    0.8    1.0
```

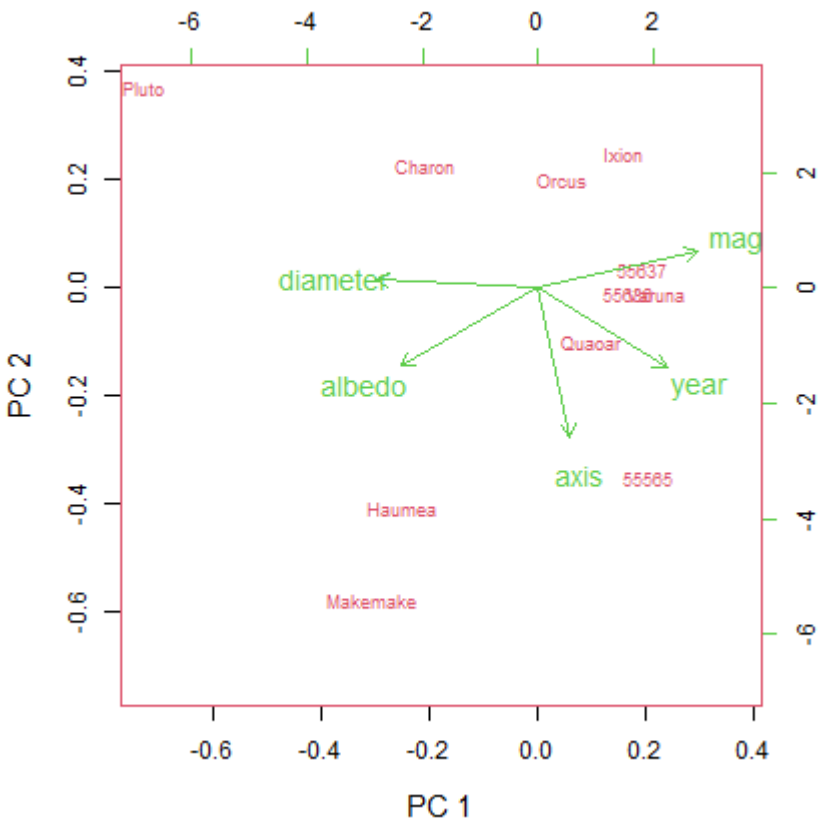
The first principal component of the Kuiper objects is approximately

$$\text{Magnitude} + \text{Year of discovery} - \text{Albedo} - \text{Diameter}$$

and the second component is almost entirely the axis.

Biplot

```
biplot(KuiPCc, col = c(2,3), cex = c(.75, 1.2), cex.lab = 1.25,
       xlabs = KBO$Permanent, xlab = "PC 1", ylab = "PC 2")
```



Summary

- It's wiser to scale the variables before PCA, or, do PCA on the correlation matrix rather than the covariance matrix.
- The principal components from the covariance matrix simply reflect the order of the sizes of the variances of the observed variables.

Other issues!

How many components?

- If you have p variables in your data, then taking all the p principal components will explain 100% variability in your data.
- The usefulness of PC stems from the idea that often the first few PCs explain most of the variability in the data.
- There are informal and formal methods for the choice of number of components. We will cover just a few popular methods, all of which are more or less *ad-hoc* in nature!

Method 1 (Proportion of total variance)

- Retain just enough components to explain some specified large percentage of the total variation of the original variables.
- Values between 70% and 90% are usually suggested, although smaller values might be appropriate as p or n, the sample size, increases.

Method 2 (Kaiser's rule)

- $\lambda > 1$, or average eigenvalue.
- When the components are extracted from the covariance matrix $\mathbf{S}$: exclude those principal components whose eigenvalues are less than the average $\sum_{i=1}^{p} \lambda_i / p$.
- Why? Since the sum of the eigenvalues is same as the sum of variances, i.e. $\sum_{i=1}^{p} \lambda_i = \sum_{i=1}^{p} \mathrm{var}(X_i)$.
- Intuition: keep those PCs that account for more variance than the average for the observed variables.

Method 2 (Kaiser's rule, contd.)

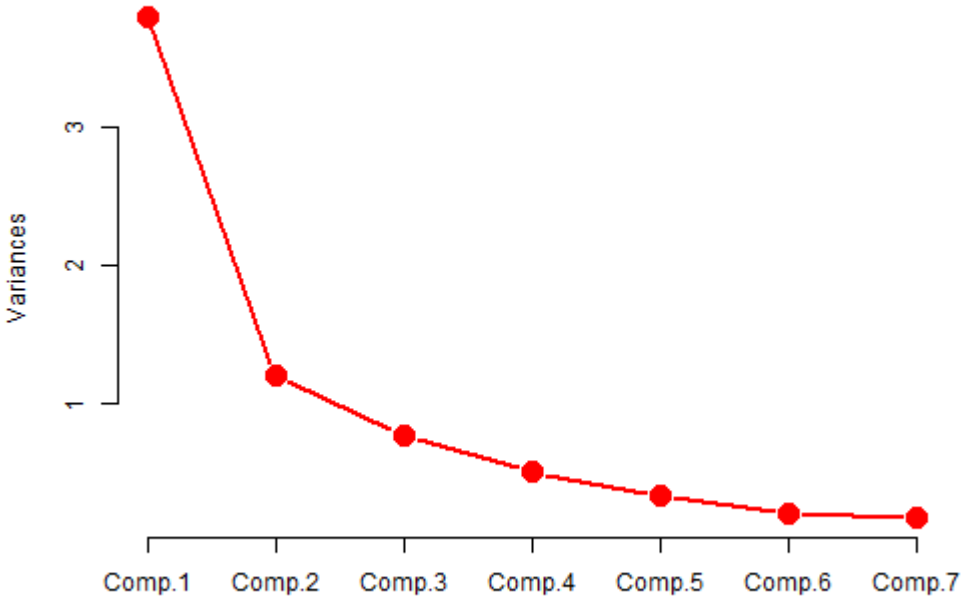
- When the components are extracted from the correlation matrix, $trace(R) = p$ = the number of variables, and the average variance is therefore $trace(R)/p = 1$.
- Applying the rule in the previous slide, components with eigenvalues less than one are excluded.
- This rule was originally suggested by Kaiser (1958), but Jolliffe (1972), on the basis of a number of simulation studies, proposed that a more appropriate procedure would be to exclude components extracted from a correlation matrix whose associated eigenvalues are less than 0.7.
- You can also apply bootstrap on Kaiser's method for estimating variance, but I'll skip that.

Method 3. Scree diagram

- Cattell (1966) suggests examination of the plot of the λ_i against i, the so-called scree diagram.
- The number of components selected is the value of i corresponding to an "elbow" in the curve, i.e., a change of slope from "steep" to "shallow".
- In fact, Cattell was more specific than this, recommending to look for a point on the plot beyond which the scree diagram defines a more or less straight line, not necessarily horizontal.

Scree diagram

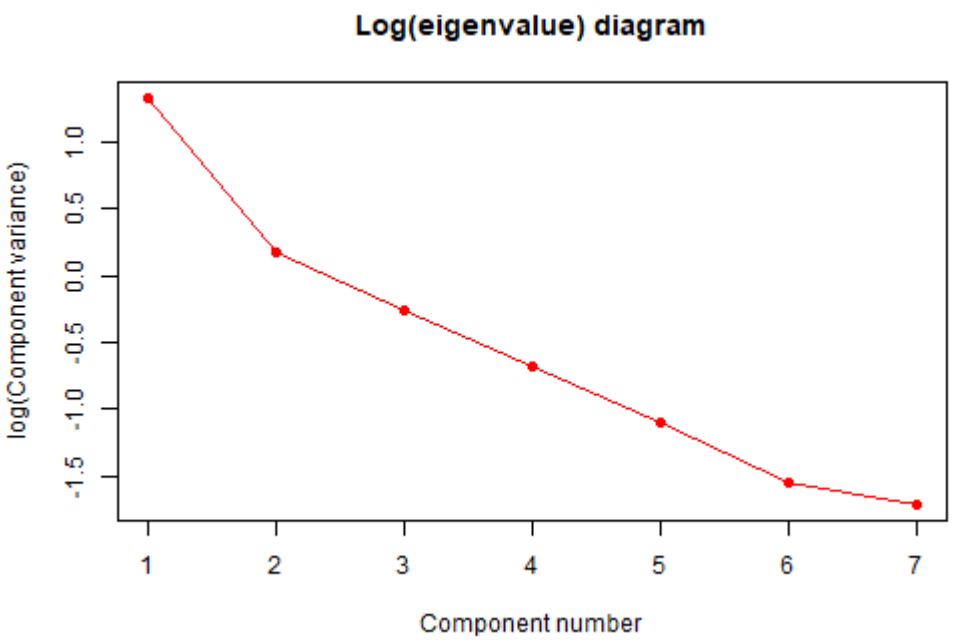
```
pca_crime <- princomp(crime, cor = TRUE)
screeplot(pca_crime, col = "red", pch = 16,
type = "lines", cex = 2, lwd = 2, main = "")
```



Method 4: log-eigenvalue plot

- Farmer (1971) proposed a modification: the log- eigenvalue diagram consisting of a plot of $\log(\lambda_i)$ against i.

```
plot(log(pca_crime$sdev^2), xlab = "Component number",
     ylab = "log(Component variance)", type="p",
     main = "Log(eigenvalue) diagram", col = "red", pch = 16)
lines(log(pca_crime$sdev^2),col = "red")
```



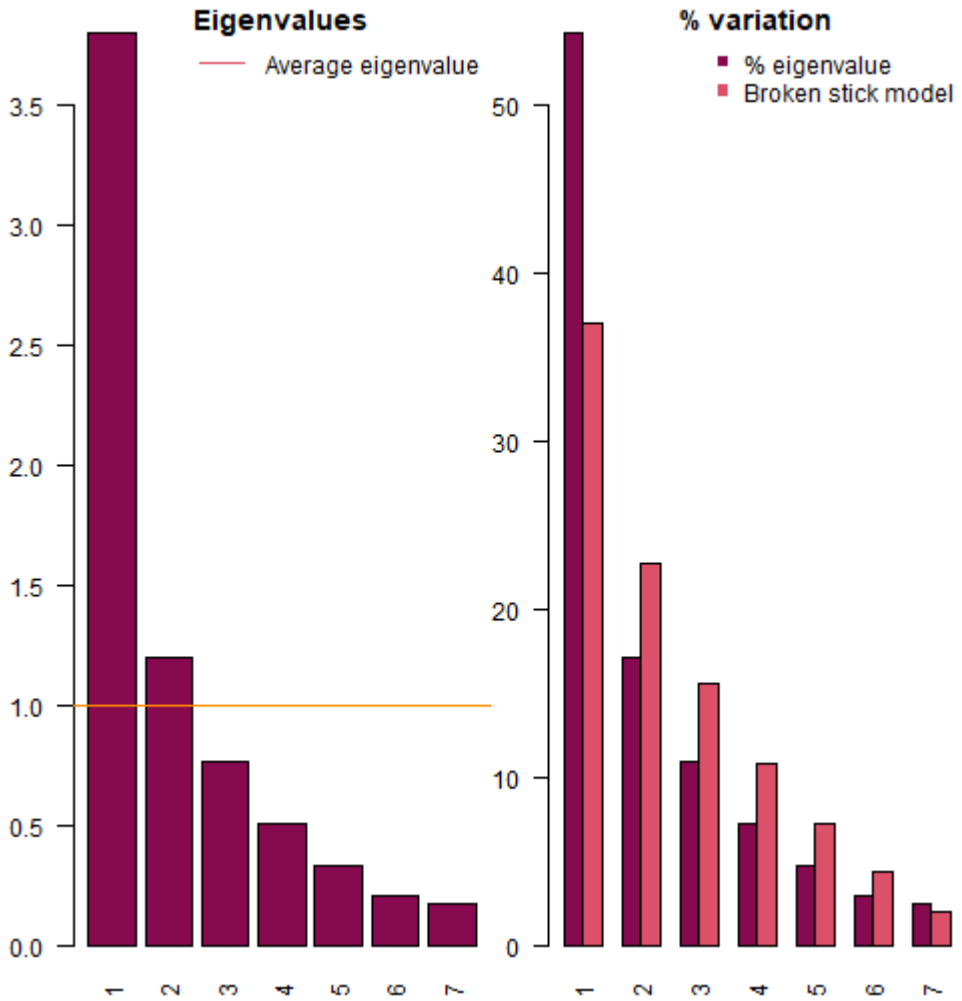
Method 5: Broken stick!

- Broken stick model
- Specifies model for eigenvalues under randomness, eigenvalue is interpretable if it exceeds the model value $\lambda_k > b_k$.

$$b_k = \sum_{i=k}^{p} \frac{1}{i}.$$

- `R` codes here: https://en.proft.me/2016/11/15/principal-component-analysis-pca-r/.

Broken stick



Best?

- Jackson (1993), Peres-Neto et al. (2005) compared various methods
- favors the broken stick model approach or bootstrap approach
- scree plot tends to overestimate (by one) the number of components
- eigenvalues >1 overestimates number

> No globally best method!
>
> Best to try to evaluate validity of smaller components.